

Introduction to Impression materials

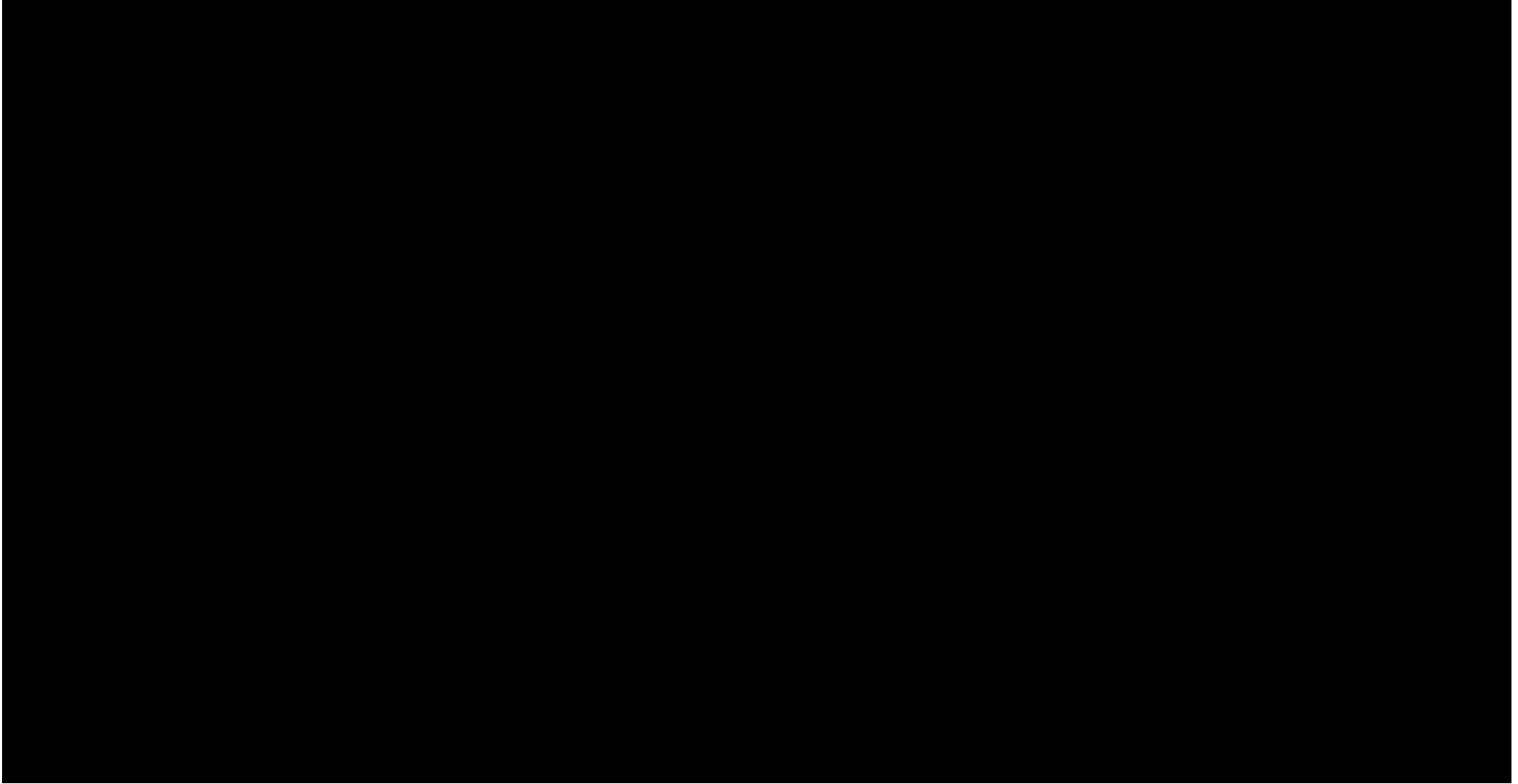
By

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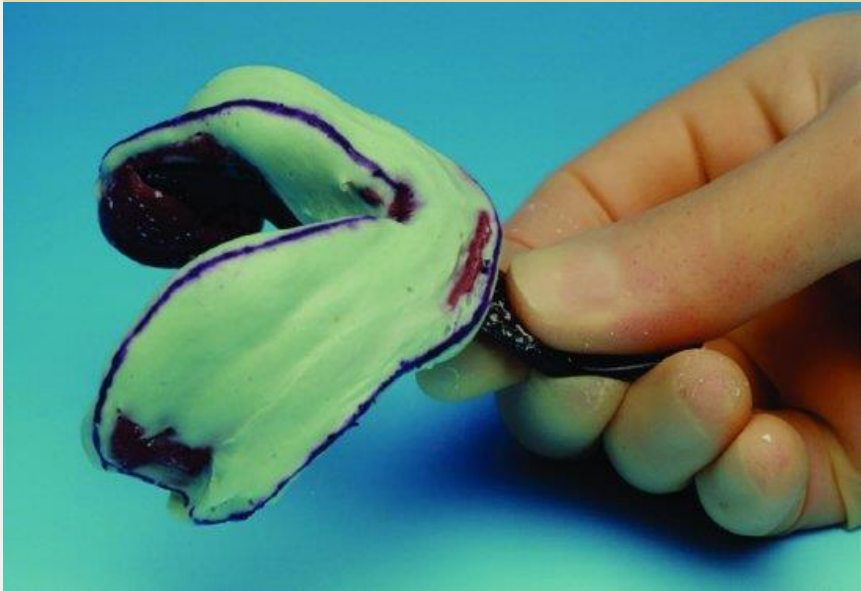


Taking impression

Practical



1-Impression materials



- **Definition:** are materials used to record the form and dimensions of oral tissues
- A material in **the plastic state** is placed against the oral tissues to set.
- After setting, the impression is removed (**elastic state**)



then poured to make a **replica of the oral tissues**.

- The impression is a **negative reproduction** of oral tissues.
- Cast is the **positive reproduction** of oral tissues.

CAST : positive reproduction takes the form of the tissues of the upper or lower jaw and for the construction of dentures, crowns, fixed dental prostheses, and other restorations, it is described as **a cast**.

The cast is made by : 1-filling the impression with dental model material (pouring)

or

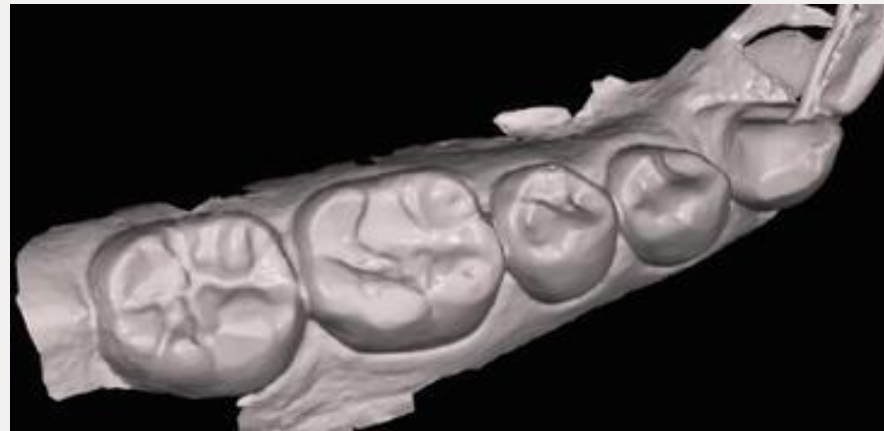
2- by scanning the impression and producing virtual cast

or

3- printing a plastic model



Cast/ model is a positive replica



Digital virtual CAST



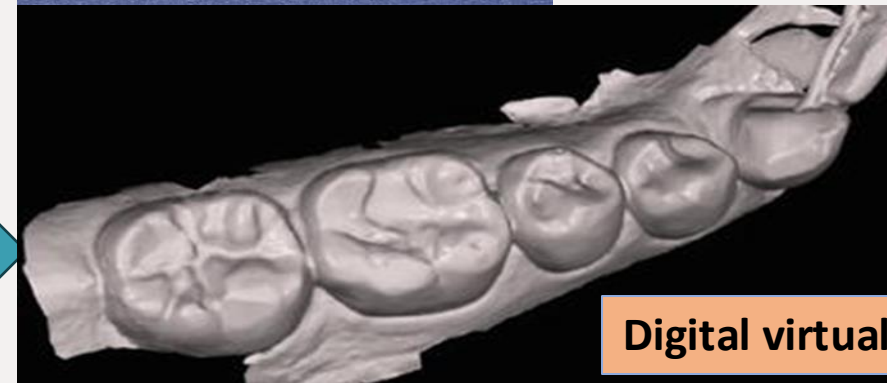
Impression

**Poured with
model
material
(gypsum)**



Gypsum Cast

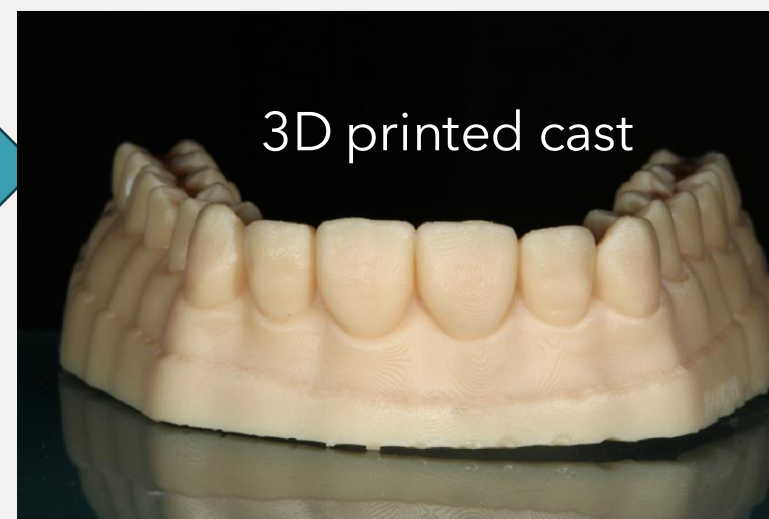
Scanning



Digital virtual CAST

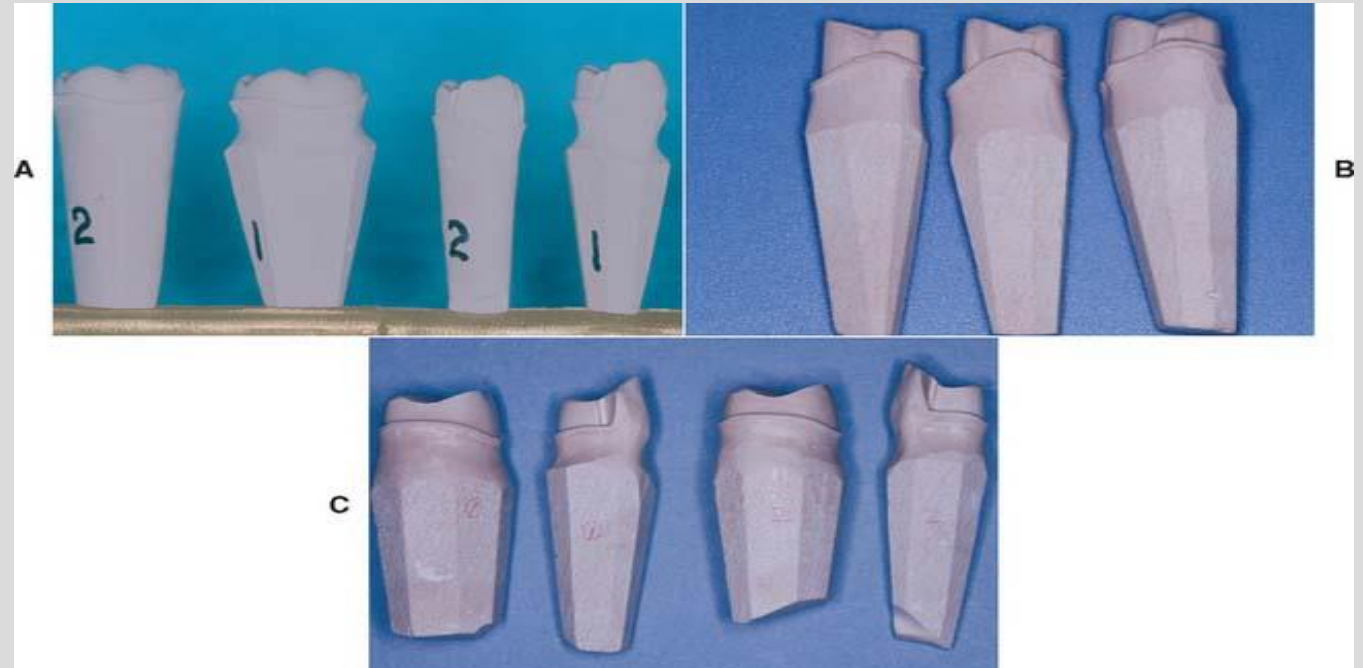
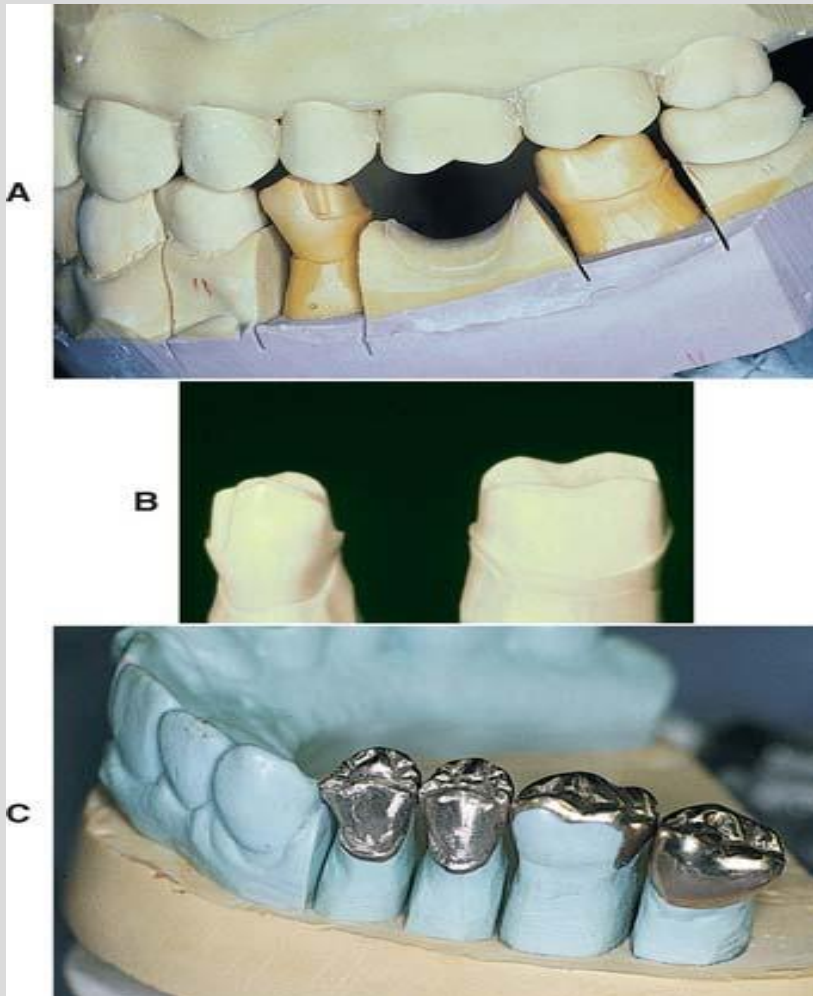


3D printing



3D printed cast

DIE: The positive reproduction of the form of a prepared tooth for the preparation of onlays , inlays or fixed dental prothesis



A to C, Examples of individual dies

<https://pocketdentistry.com/17-definitive-casts-and-dies/>

Removable die system. A, Definitive cast. B, The individual dies. C, Epoxy die

2- Trays

Impression trays are used to hold the impression material, allowing the operator to place it into the patients' mouth

Tray may be:

- 1- plastic or metal
- 2- perforated or non perforated
- 3- sectional or complete
- 4- Stock or Special (custom)
- 5- edentulous and dentulous



Practical



(a)



(b)



(c)



(d)



(e)



(f)



(g)

Practical

Metal

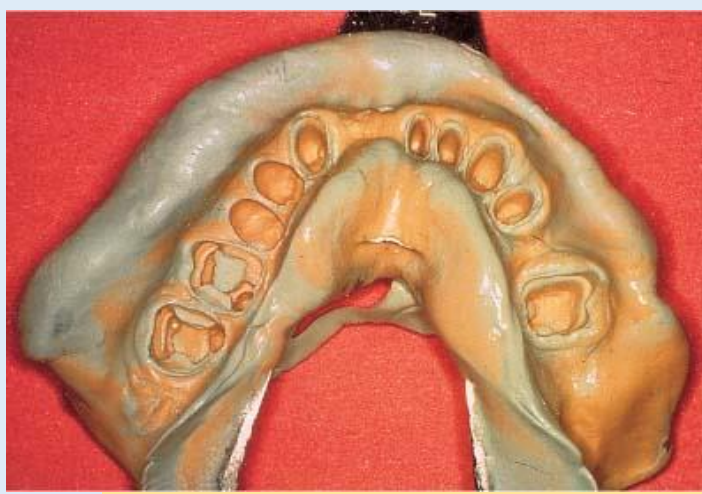


Stock tray

Acrylic resin



Special or custom tray



3- Types of impression according to accuracy and indication

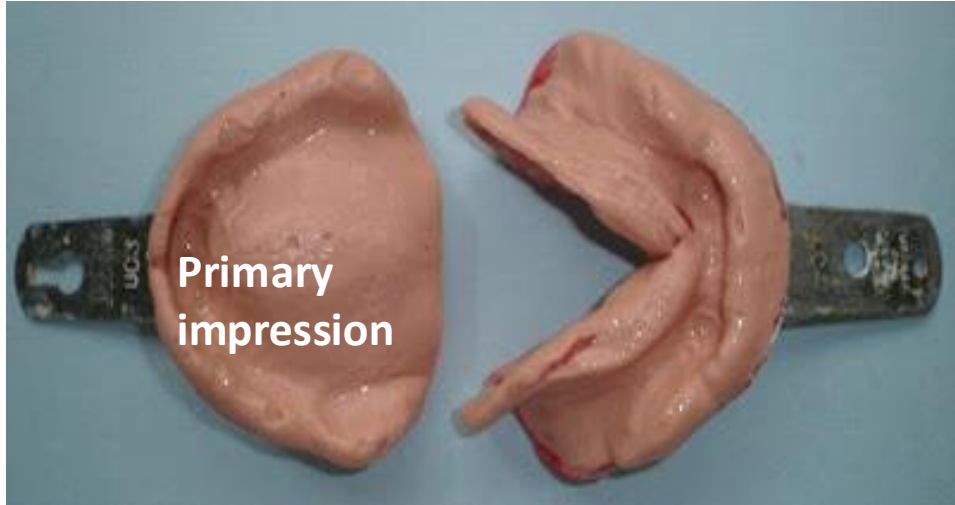


A- Primary impression materials : - as Alginate

- Using Stock Tray
- Recording Dimensions of soft and hard tissues
- Less accurate
- Pouring by model material → primary cast
- Primary Cast is used to make → special or custom tray

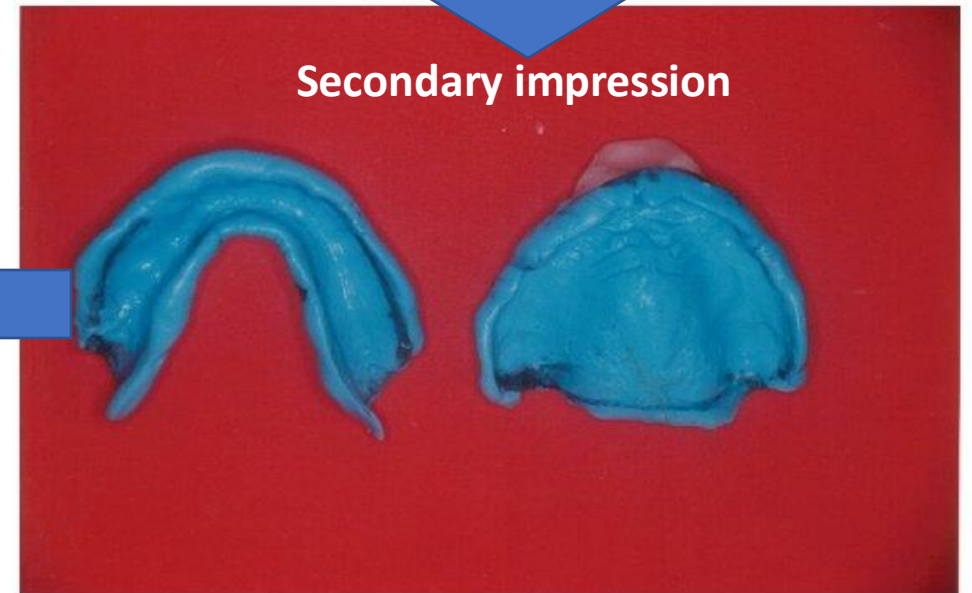
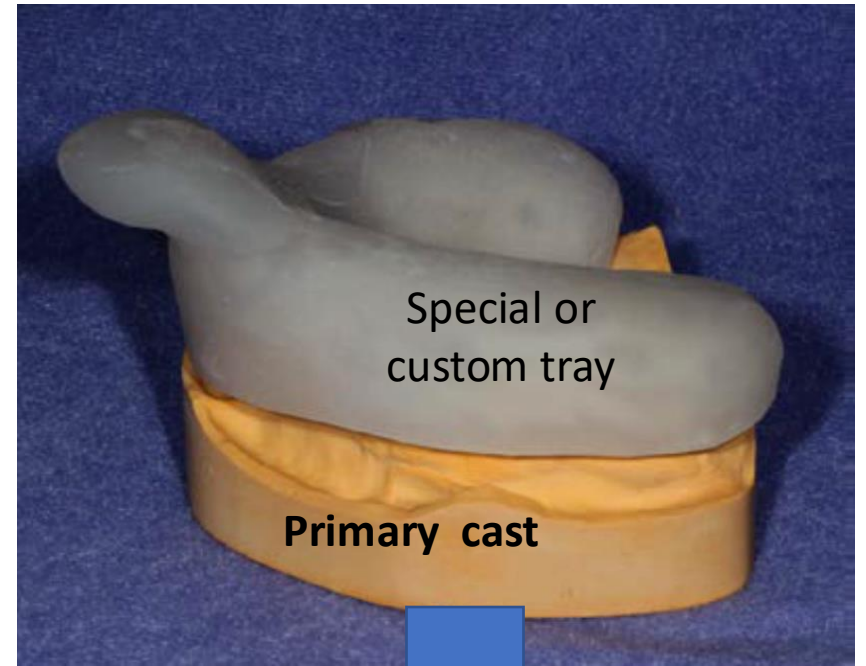
B- Secondary or Final impression: as Zn O/eugenol or Elastomers

- Using custom or special tray
- Recording fine details
- More accurate
- Pouring by model material → working cast
 - Working cast is used for → making prosthesis



<https://dentistryandmedicine.blogspot.com>

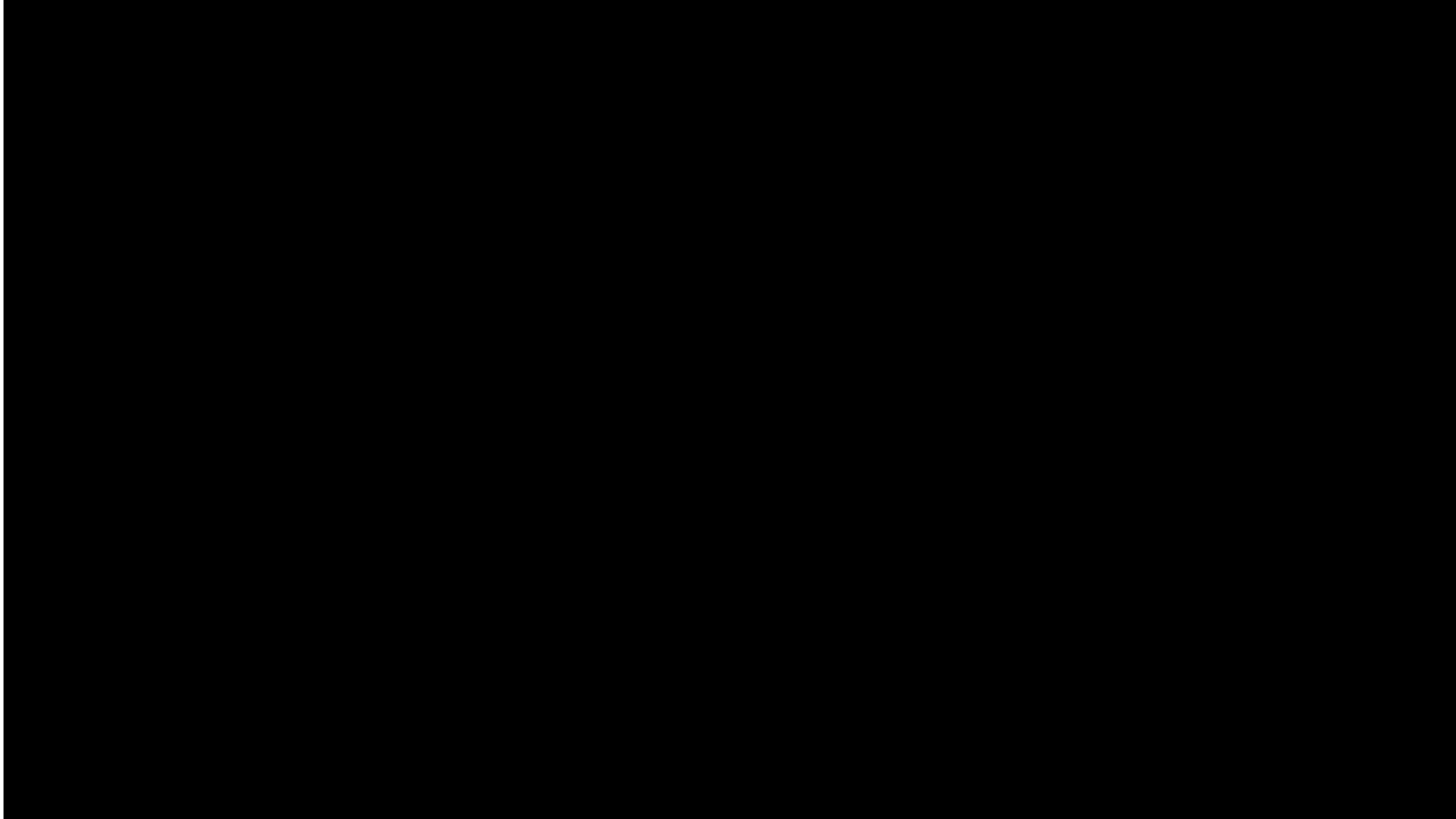
Practical



https://www.researchgate.net/publication/272370805_Clinical_Strategies_for_Complete_Denture_Rehabilitation_in_a_Patient_with_Parkinson_Disease_and_Reduced_Neuromuscular_Control/figures?lo=1

<http://www.globaldentalpro.com/2017/01/15/what-is-cast-partial-denture/>

Practical : Trays





JUST
Jordan University of Science and Technology

Lab 1: Fabrication of Maxillary and Mandibular Custom Tray

Dr. Lina Khasawneh

BDS , D.Ch.Dent Prosthodontics, FFD Prosthodontics, Jordanian Board Prosthodontics



REMOVABLE PROSTHODONTICS I
Complete denture construction
Dent 361

4- Ideal Requirements of Impression Materials

- 1• Ease of manipulation and reasonable cost e.g. Alginate
- 2• Adequate flow and wetting properties i.e. Hydrophilic
- 3 • Appropriate setting time and working time
- 4• Sufficient mechanical strength—elastic recovery and resistant to tearing during removal from undercut
- 5• Good dimensional accuracy & adequate dimensional stability





- **6• Taste and odor acceptable to patient**
- **7• Safe—not toxic or irritating i.e. Biocompatible**
- **8• No significant degradation of properties as a result of disinfection**
- **9• Compatibility with all die and cast materials**
- **10• Good shelf-life**

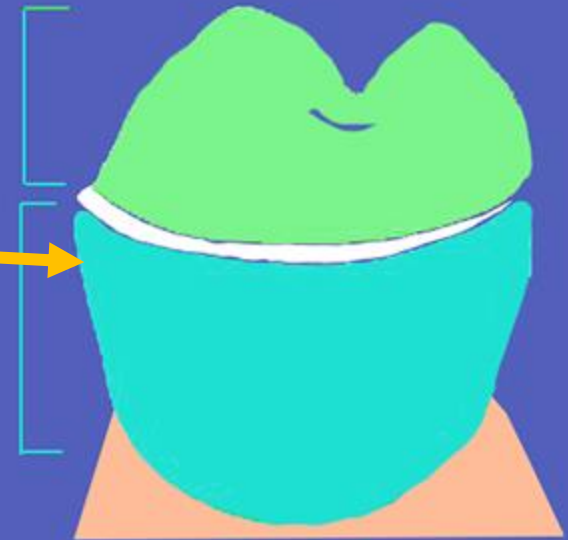
Impression materials must have high resistance to tear and high elastic recovery to record undercut areas accurately



Height of Contour

Suprabulge

Infrabulge
= undercut



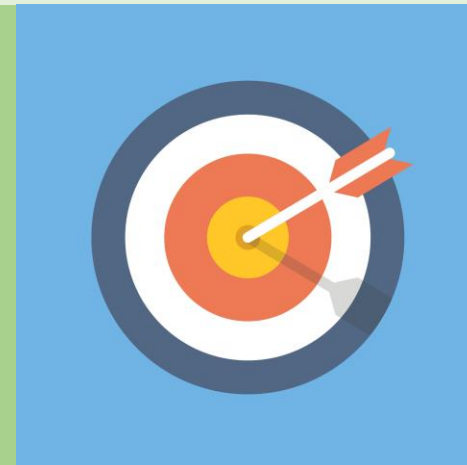
Dentulous = undercuts

<https://mydentaltechnologynotes.wordpress.com/2018/07/15/removable-partial-denture-dental-surveying/>

Dimensional stability =
is the ability to maintain
the recorded fine details
after setting and during
storage

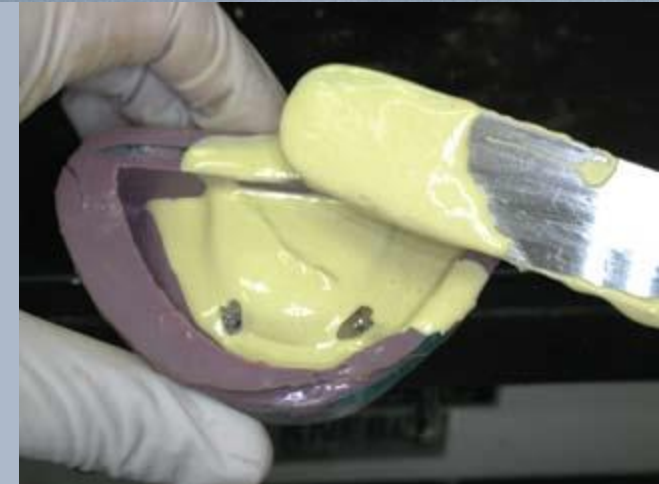


High Dimensional accuracy =
Minimal dimensional changes
during setting
(when the material converted
from the plastic to elastic state)



Compatible with model and die materials =

- Not interfere with setting of model materials
- No formation of chemical bond with model and die materials
- Hydrophilic



Working time

begins at the start of mixing and ends just before the elastic properties have developed

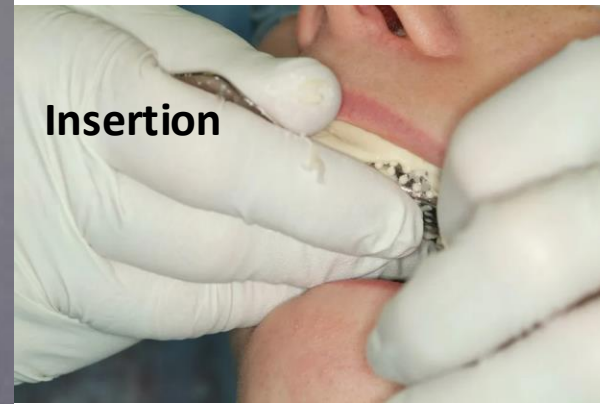
MUST BE GREATER THAN:

- the time required for mixing, filling the tray, and seating the tray.

Setting time

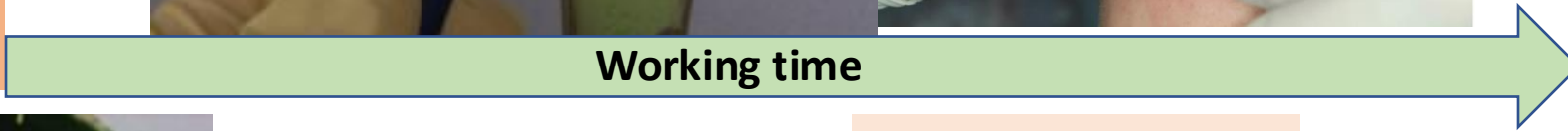
is the time that has elapsed from the beginning of mixing until the curing process has advanced sufficiently that the impression can be removed from the mouth with no distortion.

**Mixing
time**



Practical

Working time



**Loading before end
of working time**

Mixing



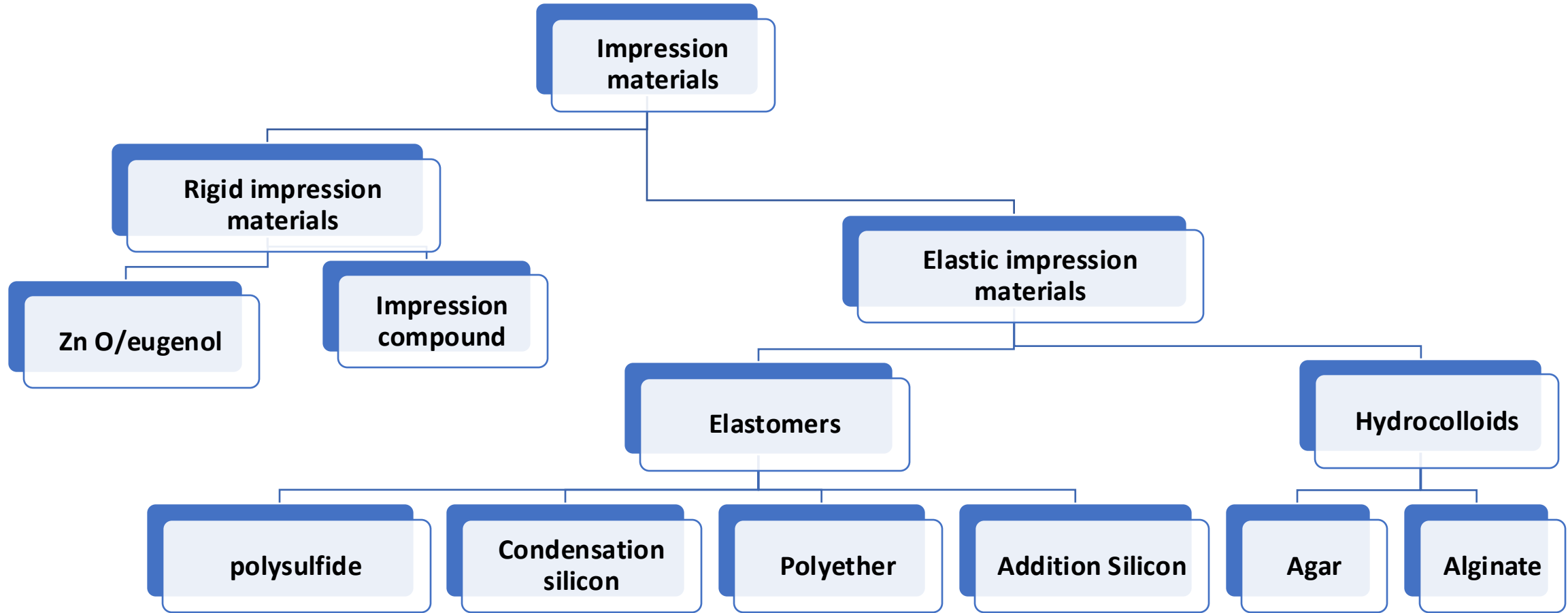
**Mixing
time**

Setting time



Removal after setting

Classification of impression materials



What is the difference between elastic and rigid impression materials?

- The impression materials after setting can be **rigid (inelastic)** or **elastic**.
- A set rigid material is highly resistant to flexure, and it fractures suddenly under stress (like upon removal from mouth), like chalk as : ZOE impression paste, impression plaster, and impression compound.
- *elastic impression materials* means that the material is flexible and can be deformed and still return to its original form to a great extent when unstressed. Examples include : agar, alginate, and elastomers.

- **Q: Can we use rigid impression materials to take impressions for intraoral areas with undercuts ?**
- **A: No, because on setting they would be locked in undercut areas (e.g. interproximal areas) and upon removal they would be permanently deformed and fractured**

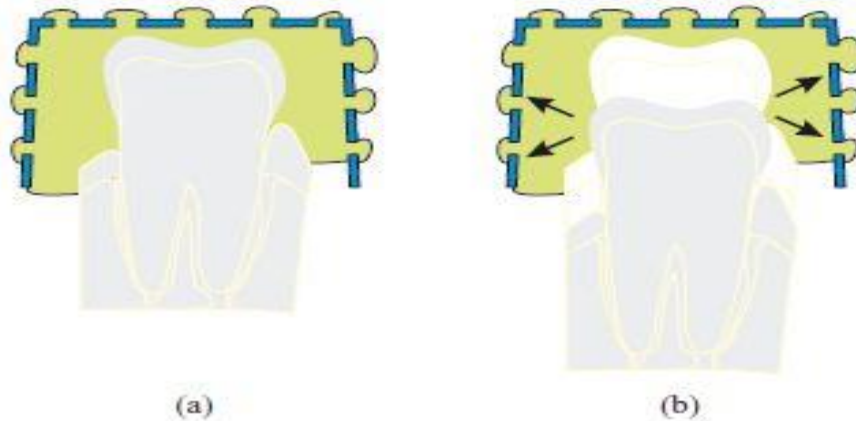


Diagram illustrating how an impression material is placed under stress during removal from an undercut area. (a) Impression in place before removal. (b) During removal



• **THANK YOU**